Data Science Pipeline
Methodology Compliance Note
Aligned to NHS Analytical Standards and UK GDPR
Week-4 (DS-3)
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1. Purpose and Scope:

This document is the formal methodology and compliance note for the **NHS Mental Health Data Science Pipeline** developed across four weeks of analytical work (1-30 May 2026). It is intended to satisfy the requirements of the **NHS England Analytical Standards**, the **UK General Data Protection Regulation (UK GDPR)**, the Data Security and Protection (DSP) Toolkit, and the Reproducible Analytical Pipelines (RAP) framework.

The pipeline ingests, processes, and analyses multiple NHS and open-source datasets to **characterise mental health prevalence**, **identify patient segments**, **detect anomalous prescribing patterns**, **forecast IAPT service demand**, **quantify geographic inequalities**, and integrate monthly MHSDS data into a master analytical store.

Intended audience: Information Governance (IG) leads, clinical governance reviewers, senior data science management, and external audit functions. This note accompanies the analytical outputs produced for Week 4 DS-3 (Final pipeline documentation).

2. Datasets:

The pipeline draws on the following datasets. All are either **openly published NHS/ONS sources** or **synthetic research datasets**. No patient-identifiable data have been processed or published.

Dataset	Source	Classification	Period	Usage in Pipeline
APMS 2014	NHS Digital	Open - Published tables	2014 survey	Prevalence baselines, distribution analysis, hypothesis testing
depression_anxiety_data.csv	Open research dataset	Synthetic	Cross-sectional	Regression modelling, K-Means clustering

mental_health_proxy_timeseries.csv	CSEW / ONS (derived proxy)	Open - Public	2002-2023	Temporal pattern detection, structural break analysis
referrals_past_data.csv	IAPT management data (aggregated)	Official Sensitive	Historical monthly	Time-series forecasting
NHS Talking Therapies - Feb 2026	NHS Digital	Open - Published	Feb 2026	Geo-spatial analysis, ICB access rates
spending-by-ccg-.csv	NHS BSA Open Data	Open - Published	Multi-year	Prescribing anomaly detection
org_details.csv	NHS ODS (derived)	Open - Published	Current	Organisation lookup for prescribing data
IoD2025 Index of Multiple Deprivation	ONS	Open - Published	2025	Geo-spatial deprivation scoring (LSOA level)
ICB_APR_2023_EN_BGC (Shapefile)	ONS Geoportal	Open - Published	Apr 2023	ICB boundary mapping
LSOA to ICB Lookup	ONS Geoportal	Open - Published	Apr 2023	Geographic join (LSOA - ICB)
MHSDS Monthly - Feb 2026	NHS Digital (auto-downloaded)	Official Sensitive	Mar 2025 - Feb 2026	Monthly MHSDS integration into master pipeline.

2.1 Data Flows:

All datasets are loaded locally from a controlled working directory (datasets). **No patient-level data leave the secure analytical environment.** The MHSDS pipeline (Week 4 DS-2) downloads directly from the NHS Digital publications page over HTTPS and stores the master CSV in the outputs directory. No data are shared with third-party systems.

Key controls in place: all input files are read-only and are never modified in place. Outputs are written to a separate outputs/ directory. **Suppressed MHSDS cells** (619,013 of 771,626 rows flagged [c]) are retained with null values and are not imputed or re-identified. No individual-level identifiers appear in any output file.

3. Alignment with NHS Analytical Standards:

The pipeline has been designed and executed in accordance with the NHS England Analytical Standards (2021) and the Government Analysis Function Standards. The sub-sections below describe how each standard is addressed.

3.1 Reproducible Analytical Pipelines (RAP):

The pipeline fully implements the **RAP principles** mandated by NHS Digital and the Office for Statistics Regulation. All nine analytical scripts are **version-controlled in Git** and executed end-to-end via a single orchestrator (run_all.py). The Week 4 DS-1 reproducibility test (16 May 2026) confirmed **all 9 scripts pass from a clean environment in 57.0 seconds** (9/9 OK). The run log (pipeline_run_20260516_143823.log) is retained as evidence.

- **Version control:** Full Git history in repository.
- **Single-command execution:** run_all.py orchestrates all scripts end-to-end.
- **Clean-environment reproducibility:** W4-DS1 test passed: 9/9 scripts, 57.0s total.
- **Peer-reviewed code:** Code reviewed by DS team across all four weeks.
- **Documented dependencies:** requirements.txt specifies all Python libraries (numpy, pandas, scipy, matplotlib, seaborn, scikit-learn, prophet, geopandas, xlrd, statsmodels, openpyxl).
- **Automated outputs:** All outputs are written programmatically; no manual steps or Excel copy-paste.

3.2 Statistical Methods:

All statistical methods are standard, peer-reviewed techniques appropriate to the data types and research questions.

Week 1 DS-1: Statistical Baseline Model (APMS 2014)

Source: APMS 2014 published tables (Chapters 2, 4, 5, 7-12). Prevalence estimates were extracted from published survey tables; **95% confidence intervals were computed using the Wilson score method**, appropriate for proportions and robust at low prevalences. Age-standardised rates were sourced directly from APMS published tables. A **composite Z-score burden index** was constructed by standardising nine disorder-level regional prevalences and averaging across disorders. Population-weighted England averages used ONS mid-year regional populations. Sample sizes ranged from N=249 (young men) to N=805 per cell.

Week 1 DS-2: Hypothesis Testing:

Source: APMS 2014 Chapter 2 tables. Three hypothesis tests were conducted on CMD prevalence data. All tests were two-sided at $\alpha = 0.05$.

Test	Variables	Statistic	p-value	Conclusion
Chi-square	Gender vs CMD	$\chi^2 = 69.28$	< 0.001	Gender significantly associated with CMD
Chi-square	Age group vs CMD	$\chi^2 = 100.54$	< 0.001	Age significantly associated with CMD
Independent t-test	Male vs Female CMD rates	$t = -2.81$	0.016	Significantly higher CMD rates in women

Week 1 DS-3: Distribution Analysis:

Source: APMS 2014 Chapter 2 tables. Descriptive distributional analyses were conducted across population segments. **CIS-R band prevalences** (0–5, 6–11, 12–17, 18+) were plotted by sex and age group. Any CMD prevalence was analysed by ethnicity, employment status, and smoking status. A temporal trend analysis covering APMS survey waves **1993, 2000, 2007, and 2014** showed the male-female CMD gap widening from 7.2 to 7.8 percentage points over that period.

Week 2 DS-1: Regression Modelling (GAD Score Predictors):

Source: depression_anxiety_data.csv. A **linear regression model** was fitted to predict GAD-7 scores from demographic and clinical features. The dataset was split 80:20 (train/test, random_state=42). Continuous features were standardised (StandardScaler); categorical features were one-hot encoded. A sklearn Pipeline was used to prevent data leakage.

Metric	Value	Interpretation
MSE (test set)	9.45	Average squared error of ~3.07 GAD points
R^2 (test set)	0.458	Model explains 45.8% of variance in GAD score
Adjusted R^2	0.419	Adjusted for number of predictors
Top positive predictor	PHQ score ($\beta = 2.93$)	Strongest driver of GAD severity
Top negative predictor	Gender: male ($\beta = -1.27$)	Men score 1.27 points lower, adjusting for other factors

Week 2 DS-2: K-Means Clustering (Patient Segmentation):

Source: depression_anxiety_data.csv. **K-Means clustering (K=6)** was applied to PHQ-9, GAD-7, age, BMI, Epworth sleepiness score, and gender. Features were standardised. The elbow method and silhouette scoring (K=4 to 6) guided cluster selection. All assignments are reproducible via random_state=42.

Cluster	Profile Name	N	PHQ Severity	GAD Severity	Key Characteristic
0	Low-Risk Male	259	Mild (4.81)	Mild (4.30)	Young, low BMI, male majority
1	Obesity-Mild Distress	56	Mild (8.75)	Mild (7.30)	High BMI (31.3)

2	Moderate Mixed+Sleep	167	Moderate (10.13)	Moderate (10.98)	Poor sleep (Epworth 8.6)
3	Older Adult Mild	61	Mild (6.62)	Mild (5.98)	Older mean age (23.9), mixed gender
4	High-Risk Severe	46	Severe (17.17)	Severe (16.15)	Highest need; priority for intervention
5	Low-Risk Female	186	Mild (4.92)	Mild (4.52)	Young, low BMI, female majority

Week 2 DS-3: Temporal Pattern Detection (CSEW Proxy Series):
Source: mental_health_proxy_timeseries.csv. Data from 2021 and 2022 were excluded due to COVID-19 survey disruption. **Three-year rolling averages** were computed (min_periods=2). **Structural breaks were tested using the Chow test**, scanning candidate breakpoints across years 2003-2018.

Variable	Test Break (2013)	Significant?	Best Break Year	Best F-stat
Car crime worry	F=57.15, p<0.001	Yes	2012	F=79.48
Violent crime worry	F=3.98, p=0.037	Yes	2005	F=9.80
Burglary worry	F=1.21, p=0.323	No	2011	F=10.62
Female feel safe	F=0.87, p=0.437	No	2012	F=6.08
Male feel safe	F=0.17, p=0.846	No	2009	F=3.17
All feel safe	F=0.41, p=0.671	No	2012	F=3.37

Week 3 DS-1: IAPT Demand Forecasting:
Source: referrals_past_data.csv. Two models were compared: **Facebook Prophet** (additive regression with automatic seasonality handling) and **ARIMA (2,1,2)** (fitted via statsmodels). The final 6 months were held out for testing. Prophet outperformed ARIMA on both MAE and RMSE.

Model	MAE	RMSE	Selected?
Prophet	5,679 referrals	6,869 referrals	Yes — lower error on held-out set
ARIMA (2,1,2)	13,512 referrals	15,496 referrals	No

Prophet 3-month ahead forecast values:

Month	Forecast	95% CI Lower	95% CI Upper
March 2026	152,424	148,135	156,540
April 2026	126,767	122,373	131,151
May 2026	136,547	132,101	141,397

Week 3 DS-2: Prescribing Anomaly Detection (Isolation Forest):
Source: spending-by-ccg-.csv merged with org_details.csv. Five per-patient and per-item features were engineered: cost_per_patient, items_per_patient, quantity_per_patient, cost_per_item, and quantity_per_item. An **Isolation Forest** (scikit-learn) was fitted with **contamination=0.05, n_estimators=100, random_state=42**. Records flagged as −1 are classified as anomalous (approximately 5% of total records, consistent with the contamination parameter). Results are stored in anomaly_flags.csv and require clinical and operational review before any action is taken.

Week 3 DS-3: Geo-spatial Analysis (IAPT and IMD):
Sources: NHS Talking Therapies activity data (Feb 2026), IoD2025 Index of Multiple Deprivation (LSOA level), LSOA-to-ICB lookup, and ICB boundary shapefile (EPSG:27700). Five **NHS Talking Therapies performance metrics** were aggregated to ICB level: referrals received, patients accessing services, reliable recovery rate, mean wait to access, and patients waiting over 90 days. IMD scores were averaged across all LSOAs within each ICB.

A **composite Unmet Need Score** was constructed using weighted normalised indicators: IMD Score (35%), Access Rate (30%), Recovery Rate (20%), and Mean Wait (15%). ICBs were classified as **underserved** where their IMD Score was at or above the 75th percentile and they also fell at or below the 25th percentile on referrals, access rate, or recovery rate.

Week 4 DS-1: Full Pipeline Reproducibility Test:

- All nine scripts were executed end-to-end from a clean environment.
- **9/9 scripts passed in 57.0 Seconds total**.
- **Run log file saved at outputs/w4_ds1_reproducibility_test**.

Week 4 DS-2: MHSDS Integration into Pipeline:
The MHSDS pipeline auto-discovers and downloads the latest monthly publication from NHS Digital, extracts and cleans the data, and appends it to a master CSV. For February 2026: **771,626 rows loaded, 619,013 suppressed rows** retained as null, master file size 203.5 MB, covering 378 unique measures across 1,149 organisations (date range March 2025 – February 2026).

4. UK GDPR and Data Governance Compliance:

4.1 Legal Basis for Processing:
All processing is conducted under **Article 6(1)(e) UK GDPR** (public task). Where special category health data are involved, processing is additionally covered by **Article 9(2)(h)** (health or social care management) or **Article 9(2)(j)** (research). The depression_anxiety_data.csv is pseudonymised/synthetic and contains no NHS numbers, names, or postcodes. All other datasets are published aggregate statistics to which individual GDPR rights do not apply.

4.2 Data Minimisation and Purpose Limitation

- **Column selection:** Only the columns required for each analysis are loaded into memory in every script.
- **No enrichment:** No additional personal data are joined at any stage.
- **Aggregated outputs:** All published outputs are aggregated at ICB, CCG, or regional level; no individual-level results are released.
- **MHSDS:** Only the primary CSV from the published ZIP is extracted; auxiliary files are discarded.

4.3 Data Security:

- **Access control:** The project repository is version-controlled with standard organisational authentication.
- **Data in transit:** MHSDS files are downloaded from NHS Digital over HTTPS only. No data are transmitted to third-party systems.
- **No hardcoded credentials:** API keys, tokens, or passwords do not appear in any script.
- **Dependency audit:** requirements.txt lists all third-party dependencies; no dependency accesses external services at runtime.

4.4 Suppression and Disclosure Control:

The NHS Talking Therapies dataset, MHSDS data, and prescribing data all contain **pre-applied suppression by the publishing body** (NHS Digital / NHS BSA). Suppressed cells in MHSDS are loaded as null values and excluded from aggregations. **No attempt is made to recover or impute suppressed values.** The pipeline does not apply additional small-number suppression to open aggregate datasets as no individual-level data are processed.

5. Data Quality:

Dataset	Known Issue	Action Taken
APMS 2014 tables	Some cells contain '_' (suppressed/not applicable)	Converted to NaN; excluded from calculations
depression_anxiety_data.csv	Rows with missing values on required features	Complete-case analysis (dropna); no imputation
CSEW proxy timeseries	2021 and 2022 missing due to COVID survey suspension	Excluded from Chow test; flagged in plots
referrals_past_data.csv	Small dataset (few monthly observations)	6-month hold-out used; forecast CIs wide and treated as indicative
NHS Talking Therapies	SubICB data aggregated to ICB; rounding in source	Aggregation method (sum/mean) documented in script
Prescribing data	Infinite values from division by zero (list size = 0)	Replaced with NaN and excluded (dropna)

MHSDS Feb 2026	619,013 of 771,626 rows have suppressed MEASURE_VALUE	Retained as null; not imputed
IMD 2025 / LSOA lookup	IoD2025 uses 2021 LSOA boundaries; older data may differ	Noted as limitation; IoD2025 and lookup are consistent

6. Limitations:

- **APMS 2014 age:** Data are over a decade old. Prevalence estimates may not reflect current population mental health, particularly post-COVID. They serve as a regional baseline only.
- **depression_anxiety_data.csv:** This open research dataset has not been validated against NHS coding systems. Cluster profiles and regression coefficients should not be applied to clinical decision-making without further validation.
- **CSEW proxy series:** Crime worry and personal safety indicators are distal proxies for mental health. No causal claim is made.
- **IAPT forecast:** Based on a short historical series. Forecast confidence intervals are wide and should be treated as indicative only.
- **Isolation Forest contamination:** The 5% contamination parameter is an assumed rate; the true proportion of anomalous CCG-months is unknown. All flagged records require clinical and operational review before action.
- **Geo-spatial analysis:** ICB-level aggregation cannot identify sub-ICB hotspots. The composite Unmet Need Score weightings were chosen by the analytical team; alternative weightings would alter the classification.
- **MHSDS coverage:** Currently covers one month (February 2026). Analyses are illustrative until further months are appended.
- **Causal inference:** No causal inference is drawn from any analysis. All associations are descriptive or predictive.

7. References:

- NHS England Analytical Standards (2021) - england.nhs.uk/publication/analytical-standards.
- UK GDPR and Data Protection Act 2018 - legislation.gov.uk.
- NHS Data Security and Protection Toolkit - dsptoolkit.nhs.uk.
- NHS Digital Reproducible Analytical Pipelines guidance - nhsdigital.github.io/rap-community-of-practice.
- Office for Statistics Regulation - Quality Statistics in Government.
- APMS 2014 - Adult Psychiatric Morbidity Survey, NHS Digital.
- Index of Multiple Deprivation 2025 - ONS.
- Crime Survey for England and Wales (CSEW) - ONS.
- NHS Talking Therapies Monthly Statistics - NHS Digital.
- Mental Health Services Monthly Statistics (MHSDS) - NHS Digital.